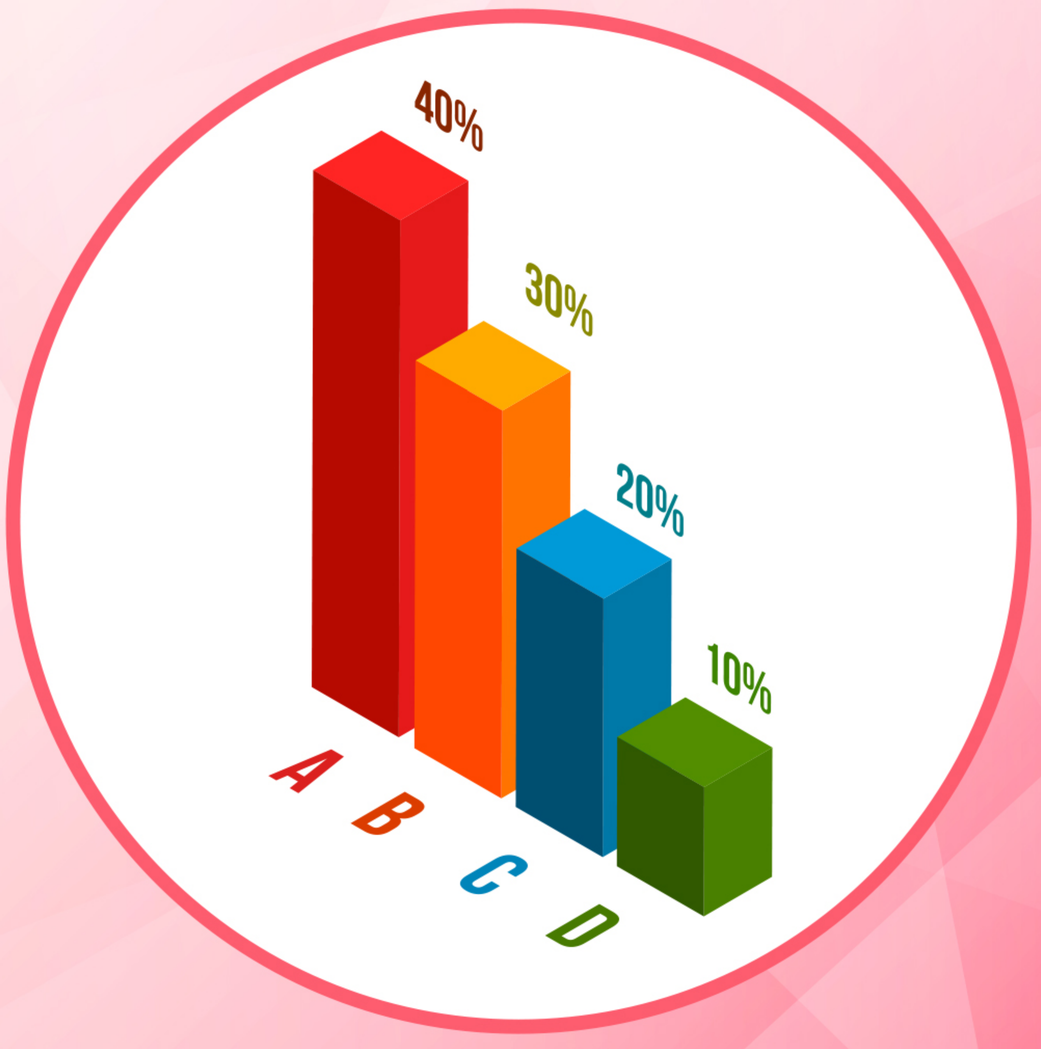
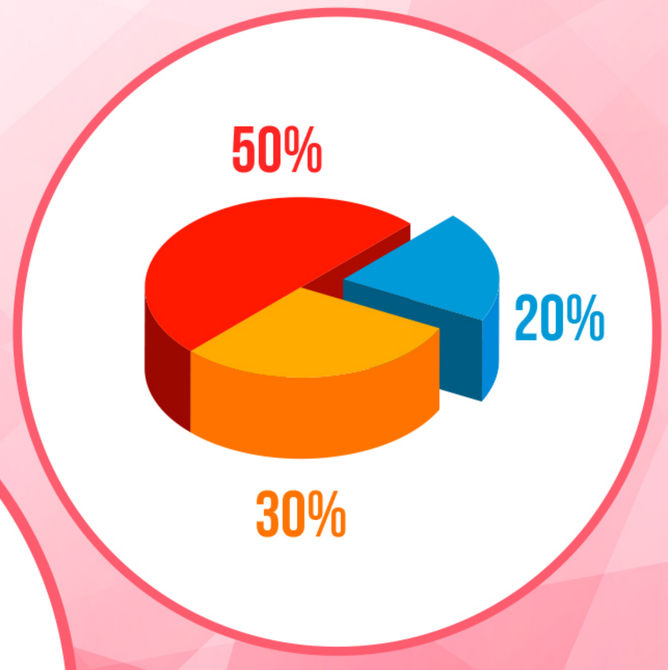
DATA

Tables & other important Concepts for DI

Assignment 03





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DI- Tables Online Assignment -03

DIRECTIONS ***for questions 1 to 5:***Refer to the following information and answer the questions that follow.

Ladooji is deciding on the formation of his cabinet after just having lead his coalition to victory in the vote of confidence. The coalition that has been patched through consists of 4 parties - the Dal, the Rasam, the CMK and the DPI. The small cabinet of 28 ministers must satisfy the following conditions
The Dal and the Rasam should have 10 ministers each
The CMK and the DPI must have 4 ministers each
Atleast 6 ministers must come from each of the 4 regions of the country - the North, South, East/Northeast, and West.
The DPI and the CMK ministers cannot be selected from the same region.
Following are the nominations received from the parties –

	Dal	*Rasam*	*CMK*	*DPI*
South	4	4	2	2
West	4	4	-	-
East / North east	4	-	-	4
North	6	2	2	2

1. Which of the following is true?

I) Six ministers cannot be selected from the north
II) DPI ministers must be selected only from the East and North east

1. I only 2. II only 3.Both I & II 4.Neither I nor II 5. Can't say

2. If 10 ministers are from the North, then how many Dal and Rasam ministers are from the south?

1. 22 2. 0, 0 3.4, 2 4.0, 4 5. 1,3

3. If 10 ministers are selected from the South, then how many Rasam ministers from the west have to be included?

1. 0 2. 2 3.5 4.6 5. 4

4. Which region can send 12 ministers?

1. North 2. South 3.Both 4.Neither 5. West

5. The probability of selection is the highest for a MP from

1. West 2. Dal 3.East and N/E 4.Rasam 5. West & Dal

DIRECTIONS ***for questions 6 to 10:***Refer to the following information and answer the questions that follow.

Given below are a set of countries and a set of indicators for each of these countries.
Two countries i and j are said to be positively correlated with respect to the indicators A and B if either (i) Ai >Aj and Bi >Bj occur simultaneously, or (ii) Ai <Aj and Bi <Bj occur simultaneously
Similarly they said to be negatively correlated if either: (i) Ai >Aj and Bi <Bj occur simultaneously, or (ii) Ai <Aj and Bi >Bj occur simultaneously. The quantity Q for any country is defined as $Q = \alpha P - \beta I + \gamma L_1$, where P, I and L_1 are respectively the per capita income, the infant morality and the literacy rate for the country and α, β and γ are unknown equal positive constants.

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Country	Per Capita Income P$/year	Per Capita Energy consmp. EKWH	Literacy L_1%	Infant Mortality I per 1000	Life Expectancy L_2 years
USA	21,800	10,800	98	8	76
Sri Lanka	3,000	300	89	22	72
Norway	20,800	6,818	100	6	77
Japan	37,000	4,700	100	4	79
Brazil	5,000	790	81	60	62
Syria	5,700	1,100	64	10	75

6. Among the four countries USA, Norway, Sri Lanka and Syria, which of these statements is certainly true?

1. Norway has a higher Q value than USA.
2. Q values follow the order USA > Syria > Sri Lanka.
3. Norway has the highest value among the four countries.
4. Q values follow the order Syria > Sri Lanka > USA
5. Q values follow the order Syria > USA > Sri Lanka

7. Given below are sets of three countries. Which one of these sets exhibit positive correlation with respect to the indicators P and E for every possible pairs of countries from the set?

1. USA, Norway & Japan
2. Brazil, Japan and USA
3. Syria, Norway, USA
4. USA, Syria&Japan
5. Norway, Japan & Syria

8. Consider only the countries Norway and Syria. Given below are sets of three indicators. Which of these sets does not show the following property: positive correlation exists with respect to one pair of indicators and negative correlation with respect to the other two pairs?

1. P, E & I
2. P, E & L_1
3. P, I & L_1
4. E, I & L_1
5. Can't say

9. If we postulate that the per capita incomes of Sri Lanka and USA double themselves every 10 years and every 20 years respectively, the per capita income of Sri Lanka will be greater than that of USA after how many years?

1. 20 years
2. 40 years
3. 60 years
4. 70 years
5. Never

10. Which of the following statements is not true?

1. USA and Sri Lanka are positively correlated with respect to P & E.
2. Syria and Sri Lanka are negatively correlated with respect to P & L_1.
3. Norway and Japan are positively correlated with respect to E and P.
4. Brazil and Norway are negatively correlated with respect to L_1 and I.
5. All of the above statements are true.

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Answer Key & Explanations

Sr. No.	Key	Explanation
1	2	Only 2nd Choice is true as min. 4 ministers have to be selected from DPI party, also DPI and CMK Ministers can't be from same region.
2	4	10 ministers must be included from Rasam party, therefore all its ministers have to be included and only 4th choice satisfies the condition.
3	5	All Rasam ministers have to be included.
4	4	No region can send 12 ministers as CMK and DPI can't be from same region.
5	4	Rasam will have highest probability of 100% as all its ministers have to be selected.
6	2	Since the constants are equal, the values of P, I & L_1 will determine the Q value in accordance with them. USA has the highest Q value as compared to Syria & Sri Lanka. Syria has higher Q value than Sri Lanka.
7	3	USA & Japan do not meet the criterion of having positive correlation with respect to P & E as P of Japan is more than USA but E of Japan is less than USA. Since 3rd option does not have both, it is the answer after choice elimination.
8	2	Indicator happens to violate the relations situation for Norway & Syria. 2nd option does not carry I indicator, thus it is the answer after choice elimination.
9	3	The Sri Lanka will become 12000 and USA will become 43600 after 20 yrs (4 times and double). After 20 more years their respective values will be 48000 and 87200. After 20 more years their respective values will be 192000 and 174400. Thus in 60 years the per capital of Sri Lanka will become more than the per capita of USA. Thus 3rd option.
10	3	1, 2 & 4 are true. Norway & Japan have inverse relations under the E & P indicators. Thus no relation can be arrived at.